iConnect Email Campaign Reporting

Investigation findings and implemented improvements | Client technical briefing

1. Overview

Following concerns about email campaign reporting accuracy within iConnect, an investigation examined the way delivery outcomes, bounces, opens and link clicks were recorded and displayed.

Two principal issues were identified: (1) temporary delivery failures could remain incorrectly labelled as bounced after successful delivery, and (2) automated email-security activity and overlapping tracking mechanisms could inflate reported clicks and opens. These were reporting and interpretation issues rather than evidence of a general failure to send campaigns.

Following the investigation, changes have been implemented to improve delivery-status processing, engagement reporting and handling of permanently rejected addresses.

2. Delivery investigation: 5 October

The following events relate to the same message sent to a recipient at Leeds. All times are UK time.

Time	Recorded event
14:34:19	Leeds' Trend Micro email gateway returned 421 - timeout exceeded. Mailgun classified the failure as temporary and scheduled a retry.
14:44:19	The retry succeeded: receiving server returned 250 - accepted and queued.
14:44:38–14:45:54	A concentrated burst of link-tracking requests was recorded against the same email.

Root cause

The synchronisation logic treated every failed event as bounced, including temporary errors, and prioritised bounced over subsequent delivered or clicked events without adequately considering their order. The earlier temporary failure therefore overrode the later successful delivery in the user interface.

Resolution implemented

Delivery-status processing now distinguishes temporary from permanent failures and takes subsequent successful events into account. A temporary failure followed by confirmed acceptance should no longer remain shown as a final bounce.

3. Soft and hard bounce handling

	Soft bounce	Hard bounce
Meaning	Temporary delivery problem	Permanent rejection of the delivery attempt
Examples	Gateway timeout, unavailable server, full mailbox	Nonexistent recipient, invalid address, permanent policy rejection
Typical provider action	Retry automatically	Normally stop retries
iConnect approach	Allow retries; do not immediately suppress	Suppress campaign sends pending review

A hard bounce does not always mean an address is misspelled: policy or security decisions can permanently reject an otherwise valid mailbox. Equally, repeated soft failures may eventually exhaust provider retries.

Implemented handling includes differentiation of failure types, correcting final outcomes after successful retries, suppression of hard-bounced addresses until reviewed, and an auditable re-enablement process. Delivery suppression remains separate from unsubscribe and consent preferences; resolving a bounce must never automatically resubscribe a recipient. Provider-side suppression must also be checked before resuming sends.

4. Unexpected click counts

The same message reported 84 clicks, but event logs show a rapid concentration of tracking requests that should not be interpreted as 84 individual human interactions.

Source / measure	Recorded finding
iConnect internal tracker	52 requests across 19 destinations
Mailgun tracking	56 click events
Concentrated activity	Approximately 76 seconds; 22 local requests in fewer than three seconds
Previously displayed figure	84 clicks - not a reconciled unique count

The timing and distribution strongly suggest automated link inspection by an email security service. The available evidence does not conclusively identify the scanner. Because both tracking paths updated the same counter, some activity could be double-counted; the exact historical total cannot reliably be reconstructed from retained records.

Resolution implemented

The reporting approach now provides improved duplicate handling, filtering of high-confidence automated link activity, separation of suspected security scanning from engagement estimates and preservation of underlying tracking records. Filtered clicks remain estimated engagement, not verified unique human interactions.

5. Email opens and privacy protection

An open event generally indicates that a tracking image was requested, not necessarily that the recipient read the email. Security scanners, image preloading and privacy features such as Apple Mail Privacy Protection can retrieve images automatically. Conversely, a person can read an email with images blocked without generating an open event.

Reporting therefore treats opens as recorded tracking-image activity rather than definitive evidence of reading.

6. Revised reporting definitions

Metric	Meaning
Delivery status	Latest confirmed delivery outcome, including successful retries.
Soft bounces	Temporary failures that may later resolve.
Hard bounces	Permanent rejections that require review.
Recorded opens	Tracking-image requests, including possible automated loading.
Filtered clicks (estimated)	Clicks after high-confidence automated activity and duplicate tracking are filtered.
Suspected automated clicks	Likely scanning activity, separately identified with supporting explanation.
Raw tracking activity	Original events retained for audit and investigation.

7. Operational benefits

- Clearer delivery outcomes: successful retries are not obscured by earlier temporary failures.
- More meaningful analytics: suspected security scanning and overlapping tracking are less likely to distort engagement figures.
- Better list hygiene: permanently rejected addresses can be reviewed and suppressed, helping protect sender reputation.
- Transparency and auditability: administrators can distinguish raw events from filtered estimates while retaining evidence for investigation.

8. Conclusion

The Leeds email examined on 5 October was successfully accepted by the receiving server after Mailgun retried a temporary timeout. The continued bounced label was a reporting error. The reported click count was also not reliable evidence of human engagement: the activity pattern strongly indicates automated inspection and possible duplication between tracking sources.

The implemented improvements address these issues and provide a more reliable, transparent basis for campaign delivery and engagement reporting. Some email engagement metrics remain inherently approximate because recipient privacy tools and corporate security systems affect what can be observed.